MedX

Every word, understood.



MedX: Multilingual AI Copilot for Clinic Visits

Research prototype · synthetic data only · Track 2 Health & Wellbeing · AMD Developer Hackathon ACT III

EXECUTIVE SUMMARY

Without a shared language, communication errors become safety events.

THE PROBLEM

49.1%

of adverse events in limited-English patients involved physical harm, vs 29.5% for English speakers.

Divi et al., Int J Qual Health Care 2007 (6 US hospitals)

OUR ANSWER

1 journey

Patient voice intake, nurse and doctor see fields linked to the patient's own words, a pharmacist verifies. People decide.

Research prototype, synthetic data only

THE PROOF SO FAR

1,872

automated tests, 18 deterministic red-flag rules, 52 synthetic cases and an 18-case blind holdout.

npm --prefix apps/web test; eval/, data/

WHO	TODAY	WITH MEDX
Patient	Cannot explain symptoms, medicines or allergies in the clinic's language	Speaks or types in Thai, Italian or English; confirms the transcript
Nurse and doctor	Free text, interpreter wait, documentation after the visit	One screen: every field quotes the patient; unknowns shown as unsupported
Pharmacist	Medication list built from memory and brand names	Reconcile list with allergy and interaction checks; verify or return



"I do the legwork. You keep the final say." People decide. MedX supports clinicians and never diagnoses.

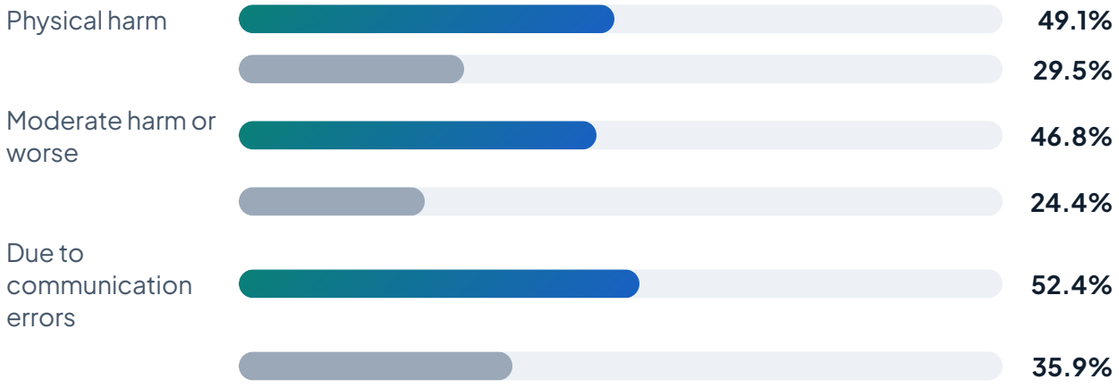
Divi C, Koss RG, Schmaltz SP, Loeb JM. Language proficiency and adverse events in US hospitals: a pilot study. Int J Qual Health Care 2007;19(2):60–67. PMID 17277013. Test and rule counts: repo, 18 Oct 2026. Not clinically validated.

THE PROBLEM: LANGUAGE BARRIER

Limited English proficiency doubled the odds of an adverse event for hospitalised children.

SIX US HOSPITALS, 2005

Share of adverse events: limited English proficiency (LEP) vs English speakers



Colour bar = LEP patients, grey bar = English speakers. Divi 2007, PMID 17277013.

7-HOSPITAL COHORT, 1,666 PARENTS

17.7% vs 9.6%

adverse events in children of parents with limited comfort in English (adjusted OR 2.1, 95% CI 1.2–3.7).

JAMA Pediatrics, Dec 2020. PMID 33074313

DELAY AND UNDER-ASSESSMENT

33 studies

in a pediatric-surgery scoping review: LEP patients had fewer pain assessments, longer waits for analgesia, longer time to treatment.

Liu O et al., J Surg Res 2024;302:540–554. PMID 39178570

WHAT HELPS

28 studies

in a systematic review: professional interpreters are linked to better communication, outcomes and satisfaction than ad hoc interpreters.

Karliner LS et al., Health Serv Res 2007;42(2):727–754. PMID 17362215

HONEST CAVEAT

Length-of-stay effects are inconsistent across 11 studies (2025 scoping review). We do not claim that language barriers lengthen stays.

Kulkarni S et al., BMJ Open Qual 2025;14(4). PMID 41314683



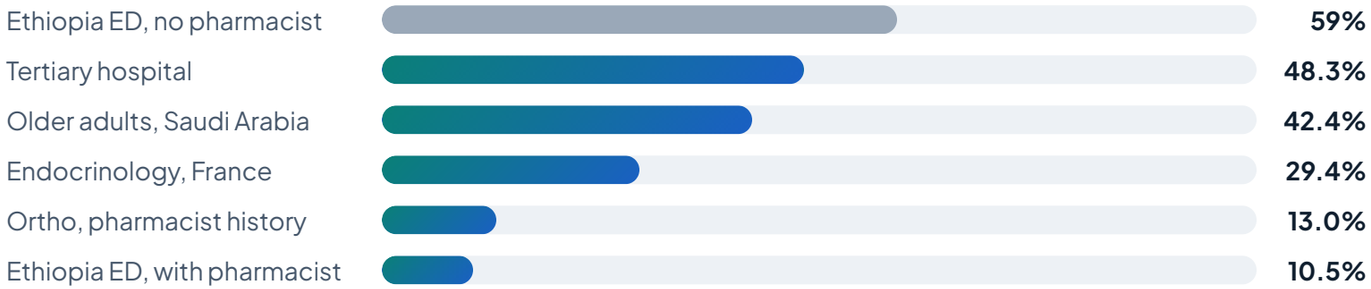
“A missed word at the desk can become a missed safety signal.” We use only figures we could trace to a paper. The rest is left out.

Sources: Divi 2007 (PMID 17277013); JAMA Pediatr 2020 (PMID 33074313); Liu 2024 (PMID 39178570); Karliner 2007 (PMID 17362215); Kulkarni 2025 (PMID 41314683). US data; no verified Thai or Italian rate was found. Full list: docs/research/PROBLEM.md.

THE PROBLEM: MEDICATION LISTS

13% to 59% of patients have an unintended error in their admission medication list.

PATIENTS WITH AT LEAST ONE UNINTENDED DISCREPANCY



Settings and methods differ, so we show a range, not an average. Order by rate. Sources M1–M6 below.

FINDING	NUMBER
Most common discrepancy type	Omission (37% in older adults, 77% of discrepancies in the tertiary study)
Could cause moderate to severe discomfort	52% of discrepancies (tertiary study)
10 or more drugs	4 times the odds of an admission discrepancy (OR 4.0)

LANGUAGE MAKES IT WORSE

13 studies

in a 2025 systematic review: low health literacy and limited English were linked to more comprehension errors and medication discrepancies.

J Racial Ethn Health Disparities 2025. PMID 40180697

WHAT MEDX DOES

Shows what the patient **said** they take, in their words, maps brand names to generics, and flags duplicates. The pharmacist confirms what they actually take.

NOT CLAIMED

We found no verified Thai or Italian error rate. This is evidence from other countries.



“A medicine the patient forgot to mention is a risk nobody can see.” That is why every field points back to what the patient actually said.

[M1] Int J Clin Pharm 2018, PMID 30367379. [M2] Int J Clin Pharm 2017, PMID 29248986. [M3] J Patient Saf 2021, PMID 28877049. [M4] BMC Health Serv Res 2026, PMID 42001177. [M5] BMC Health Serv Res 2020, PMID 32791965. [M6] Int J Clin Pharm 2018, PMID 30155773. [L1] PMID 40180697. Details: docs/research/PHARMACY-PROBLEM.md.

THE PROBLEM: ALLERGIES AND BRAND NAMES

Allergy labels and brand names travel badly across borders.

MATERNITY COHORT, AUSTRALIA, 145 PATIENTS

36%

of records with a beta-lactam allergy needed amending (52 patients). 56% of the allergies were judged low risk.

Aust N Z J Obstet Gynaecol 2024. PMID 38721775

HONG KONG COHORT, 141 ADULTS

83%

were delabelled after structured testing (117). Yet 17% of 88 followed for 12 months were incorrectly relabelled (15).

Ann Allergy Asthma Immunol 2026. PMID 41763604

CASE REPORT, NOT A PREVALENCE STUDY

A cruise passenger took a 1000 mg paracetamol cold product without knowing that paracetamol is another name for acetaminophen (Tylenol), risking an unnoticed overdose. A Reactine bought in Italy also contained pseudoephedrine.

ISMP Canada, Safe Medication Use newsletter 5(5), 18 Aug 2014

SAME INGREDIENT, DIFFERENT NAMES	WHY IT MATTERS
Tachipirina (IT), Efferalgan, Sara, Tiffy (TH), Tylenol (US)	A traveller can take two products with the same paracetamol
Lay terms: "ยาความดัน", "la pastiglia per la pressione", "my blood thinner"	The record needs a generic name, strength and last dose

WHAT MEDX DOES

Shows the allergy the patient states, **in their words**, with the reaction if given. Nothing is marked verified until the pharmacist confirms. We do not claim to remove false allergy labels.



“Brand names change at the border. The ingredient does not.” MedX maps names to generics and flags duplicate paracetamol; the pharmacist decides.

[A1] PMID 38721775 (2024). [A2] PMID 41763604 (2026). [T1] ISMP Canada, https://www.safemedicationuse.ca/newsletter/newsletter_Travelling.html. No study quantifying traveller brand confusion was found; no rate is given. docs/research/PHARMACY-PROBLEM.md.

THE PROBLEM: TIME AND DOCUMENTATION

For every hour with a patient, doctors spend almost two more hours on EHR and desk work.

57 PHYSICIANS, 4 SPECIALTIES, 430 HOURS OBSERVED



Share of the office day. Sinsky C et al., Ann Intern Med 2016;165(11):753–760. PMID 27595430

WARD NURSES, WESTERN AUSTRALIA, 120 HOURS OBSERVED

28.1% morning · **22.7%** afternoon · **20.9%** night

of time spent on documentation. Nurses estimated 50.4%, 40.7% and 37.9%: perceived burden is higher than measured, but the measured share is still large.

J Clin Nurs, Mar 2021. PMID 33590554

WHAT THIS MEANS FOR INTAKE

A patient who needs repeated clarification, or an interpreter wait, adds to a load that is already high. If the nurse starts with a structured, evidence-linked summary, the documentation starts from the patient's words instead of a blank page.

LIMIT WE STATE OPENLY

These are ward and clinic data, not triage-desk data. We found no verified triage-specific timing figure, so we claim **no time saving**. Our demo scenario of 2 to 3 minutes per intake is a design assumption, not a result.



“Less typing, more listening. That is the hope, and it has to be measured.” A measured pilot is the gate before any time-saving claim.

Sinsky C et al., Ann Intern Med 2016 (PMID 27595430); Is nursing and midwifery clinical documentation a burden? J Clin Nurs 2021 (PMID 33590554). docs/research/PROBLEM.md section 2.

THE PROBLEM: AI SUMMARIES

Good clinical LLMs still omit or invent content, so every claim needs a source.

FRAMEWORK STUDY, 12,999 SENTENCES

1.47%

hallucination rate, with a 3.45% omission rate, across 18 configurations.

npj Digit Med, May 2025. PMID 40360677

30 SYNTHETIC DUTCH LETTERS, 31 CLINICIANS

78%

judged correct. Medication sections had the highest hallucination rate; missing information was the commonest error.

JMIR Med Inform, May 2026. PMID 42114040

100 HOSPITAL-COURSE SUMMARIES

25%

had omissions and 20% inaccuracies; 2% hallucinations. 88% of unedited summaries were rated no harm potential.

JAMA Netw Open, May 2026. PMID 42101844

RISK IN CLINICAL AI	WHAT MEDX DOES ABOUT IT
Model invents a symptom or medicine	A field with no verified patient quote is `unsupported` and never shown as fact
Model misses a warning sign	Red-flag rules are code, run before any model call, and no model output can clear them
Model is unsure or fails	Low confidence, provider error or schema error means abstain: `needs human review`
Nurse trusts a field at a glance	Each field carries its quote, a confidence mark and a status; nothing is final until a clinician approves



“Show me the sentence it came from.” Low error rates are still too high when a nurse may trust a field at a glance.

[L1] npj Digit Med 2025 (PMID 40360677); [L2] JMIR Med Inform 2026 (PMID 42114040); [L3] JAMA Netw Open 2026 (PMID 42101844). Studies are on other tools and settings; they motivate our design and say nothing about MedX accuracy. docs/research/PROBLEM.md section 4.

CONTEXT: THAILAND AND ITALY

Our demo languages sit where patients and clinicians often share none.

THAILAND

FOREIGN TOURIST ARRIVALS, 2025

32,974,321

Every one of them can end up at a clinic in a language the staff may not share.

Ministry of Tourism and Sports, via Bernama, 6 Jan 2026

UNREGISTERED MIGRANT WORKERS

~338,000

from Cambodia, Laos, Myanmar and Vietnam (Department of Employment estimate, Oct 2025).

Khaosod English, 6 Oct 2025

ITALY

FOREIGN RESIDENTS, 1 JAN 2025

5.37M

9.1% of residents (ISMU report built on ISTAT data; ISTAT first release: 5.42M, 9.2%).

ISMU 31st Report on Migrations, Feb 2026. Secondary source: check against ISTAT

EUROPEAN PAEDIATRICIANS NAMING CULTURAL OR LANGUAGE FACTORS AS THE TOP BARRIER

90%

yet only 37% had access to professional interpreters and cultural mediators (n=492; Europe, not Italy-only).

Carrasco-Sanz A et al., Child Care Health Dev 2018. PMID 29159977

WHAT STUDIES IN THAILAND FOUND

987 migrant workers surveyed: language barrier (via acculturative stress) was linked to poorer access to antenatal and postpartum care.

Holumyong C et al., J Pregnancy 2018. PMID 29686905

Along the Thai-Myanmar border, communication barriers were a main factor in satisfaction with the migrant child immunisation service.

Pinna C et al., Vaccines 2020;8(1):68. PMID 32033393

LEFT OUT ON PURPOSE

Medical-tourism patient counts (sources disagree) and Italian tourist totals (definitions differ).



“Thai, Italian, English: three languages, one desk.” We start where we can demo it and where the need is documented.

Bernama 6 Jan 2026 (bernama.com/en/news.php?id=2509419); Khaosod English 6 Oct 2025; ISMU 31st Report Feb 2026 / ISTAT; Carrasco-Sanz 2018; Holumyong 2018; Pinna 2020. Tourist arrivals are a population-at-risk figure, not a count of patients.

WHO SUFFERS TODAY

One pattern: the right information is lost on the way to the person who decides.

PATIENT

Cannot say it. Symptoms, medicines and allergies get lost or simplified. Waits for an interpreter or relies on a relative.

Evidence: 49.1% vs 29.5% physical harm; 2.1× odds of adverse events (children).

NURSE

Cannot hear it all. Takes history, checks vitals and documents, often in a second language.

Evidence: 21% to 28% of shift on documentation (measured).

DOCTOR

Cannot verify it. Reads a summary with no trail back to what was said. Nearly 2 hours of desk work per face-time hour.

Evidence: 49.2% desk vs 27.0% face time.

PHARMACIST

Cannot trust the list. Brand names, lay terms and missing allergies.

Evidence: 13% to 59% admission lists with an unintended error.

One pattern. When something is lost between the patient's words and the final decision, the person at the end sees a clean-looking record with no trail back. MedX keeps the trail: every field quotes the patient, and every action is in the audit log.

Buyer view (assumption for discussion): private hospitals and clinics with international patients feel this as interpreter cost, documentation time and medication-error risk. We have not validated these costs.



“I keep the trail from the patient's words to the decision.” That is the whole product in one sentence.

All numbers are from the previous slides (Divi 2007; JAMA Pediatr 2020; J Clin Nurs 2021; Sinsky 2016; PHARMACY-PROBLEM.md M1–M6). Business statement is an assumption, not a finding.

INTRODUCING MEDX

Say it in your language. The clinician reads it in theirs, and can check every line.

WITHOUT MEDX, A TYPICAL INTAKE

1. **Patient and clinic find a common language**
   Interpreter, relative, or a phone app
2. **Nurse takes history by hand**
   Free text, written from memory
3. **Vitals on a separate screen**
   Triage level decided by eye
4. **Doctor re-asks the same questions**
   No trail to what was said
5. **Prescription typed, checked in the head**
   Allergy or duplicate seen late
6. **Pharmacist rebuilds the medicine list**
   From brand names and memory

WITH MEDX, THE SAME JOURNEY

1. **Patient speaks or types in Thai, Italian or English**
   Confirms the transcript before sending
2. **Nurse sees a live summary**
   Each field quotes the patient; unknowns say so
3. **Vitals give a NEWS2 suggestion**
   The nurse confirms the level
4. **Doctor sees considerations and a draft note**
   From a fixed list, with guideline links; signs or edits
5. **Orders are checked by rules**
   Allergy, interaction, duplicate, pregnancy, dose range
6. **Pharmacist verifies or returns**
   Reconciled list, written reason to override

ONE SENTENCE

A multilingual intake that gives clinicians a case where every field points back to the patient's own words.

WHAT STAYS HUMAN

Diagnosis, urgency decision, orders, signing and verification. The AI never decides these.

WHERE IT RUNS

Web app for staff, Expo app for patients. Live demo runs on AMD-hosted models through the AMD developer API: Gemma-4-31B-IT (red-flag screen, questions) and DeepSeek-V4-Flash (extraction).



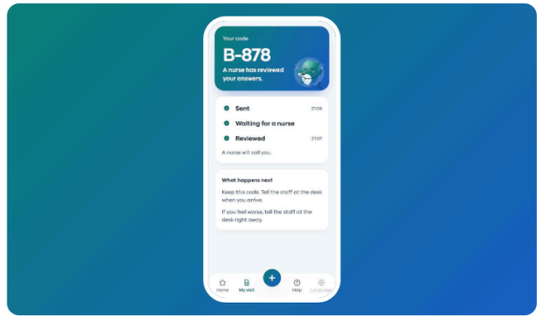
“I do the legwork. You keep the final say.” Six steps either way. With MedX each one leaves a trail.

Process steps are a design description, not a measured comparison. Not clinically validated. Research prototype, synthetic data only. See docs/DEMO.md.

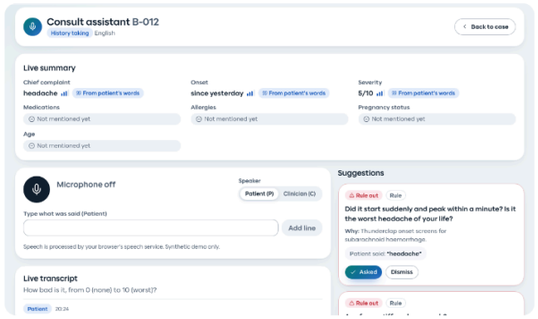
THE 4-ROLE JOURNEY

One case, four people: each gets a screen made for their job.

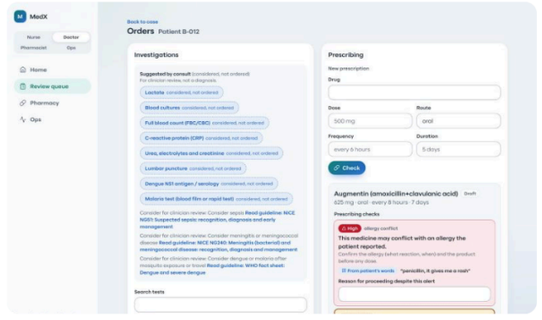
1 Patient
Speaks or types; sees when a nurse has read it



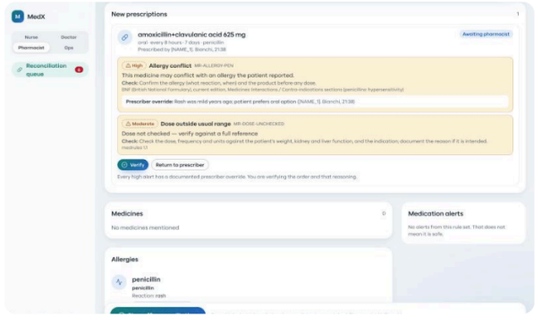
2 Nurse
Consult assistant: ask-next popups, source quotes



3 Doctor
Orders with an allergy alert and a written override



4 Pharmacist
Verify or return, with the doctor's reason in view



1 PATIENT APP

Voice or text intake

Thai, Italian, English. Guided chat, follow-up questions, red-flag screen that says go to the desk.

2 NURSE

Consult assistant + vitals

Live summary linked to the patient's words; ask-next suggestions; NEWS2 triage suggestion to confirm.

3 DOCTOR

Considerations, note, orders

Doctor-only list with guideline links; draft note to sign; orders with deterministic checks.

4 PHARMACIST

Verify or return

Reconciled medicines and allergies; verify, edit, or return with a written reason.



"Four screens, one trail." Every step is in the append-only audit log, and nothing is final until a clinician approves.

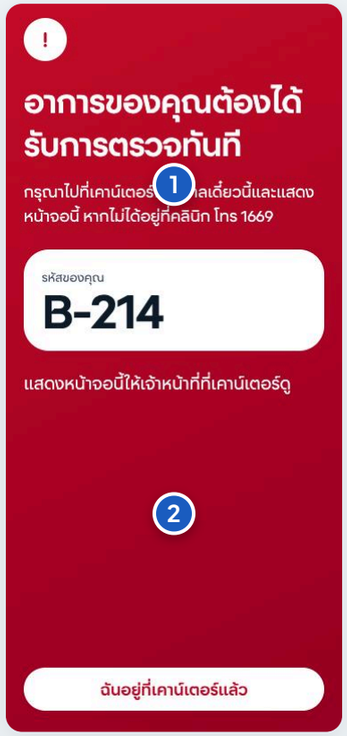
Screenshots: real app, synthetic data, mock provider (docs/readme/journey.png). Research prototype.

USE CASE: PATIENT APP

The patient answers in their own language, by voice or text, and confirms what was heard.







1. **Speak or type**
   Voice goes through the phone's speech service; text works offline of speech. The patient sees the transcript and confirms before sending.
2. **Guided questions**
   A short chat in Thai, Italian or English. If something required is missing, one follow-up question, at most 3.
3. **Red-flag screen**
   If a warning sign is detected, the app stops and tells the patient to go to the desk now. A rule fired, not a model.
4. **Redaction**
   Names, phone and ID numbers become tokens before anything is stored or sent to a model.

LIMITS WE STATE

Speech goes to the browser or phone cloud speech service. Redaction is imperfect. Italian and Thai UI wording needs native-speaker review.

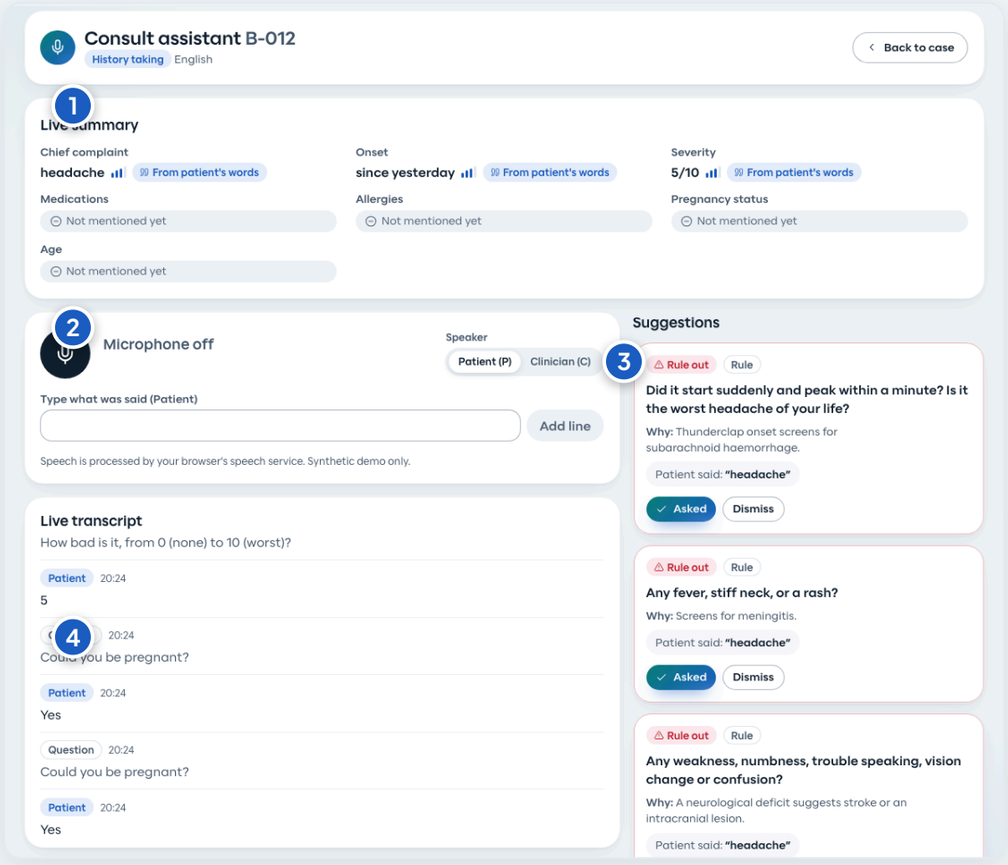


“Tell me in your own words. I will make sure the nurse sees them.” Left to right: voice intake, status, red-flag screen.

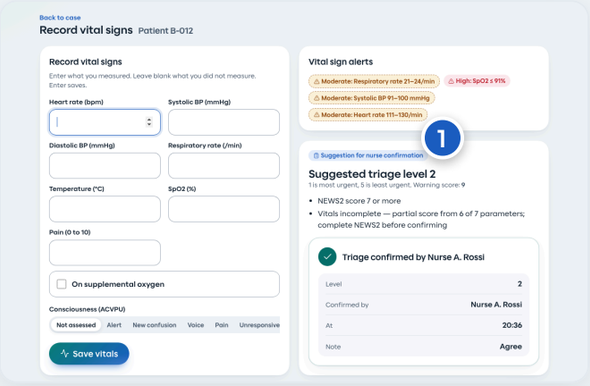
Screens: apps/mobile, synthetic data (pitch/deck/img/m-*.jpg). Follow-up cap and redaction: docs/ARCHITECTURE.md steps 2 and 8.

USE CASE: NURSE CONSULT ASSISTANT

While the nurse takes history, the summary builds itself, and every field quotes the patient.



1. **Live summary**
   Chief complaint, onset, severity. Each field shows **From patient's words**; empty ones say **Not mentioned yet**.
2. **Capture by voice or typing**
   Speaker toggle (patient or clinician). Speech goes to the browser's speech service.
3. **Ask next and rule out**
   Rule-based suggestions with a one-line reason. The nurse asks, dismisses, or marks as asked.
4. **Live transcript**
   Each line tagged patient or question; the quotes behind the fields come from here.
5. **Vitals and NEWS2**
   Seven parameters give a triage suggestion; incomplete vitals say so. The nurse confirms the level.

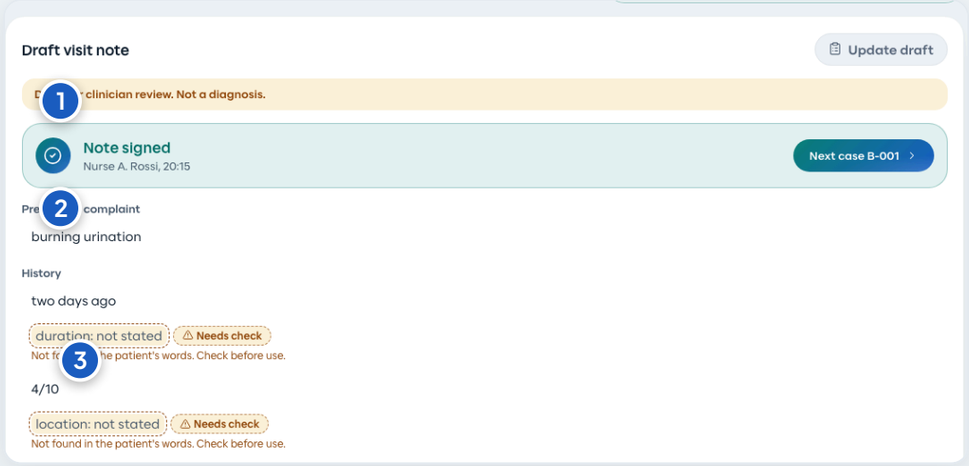


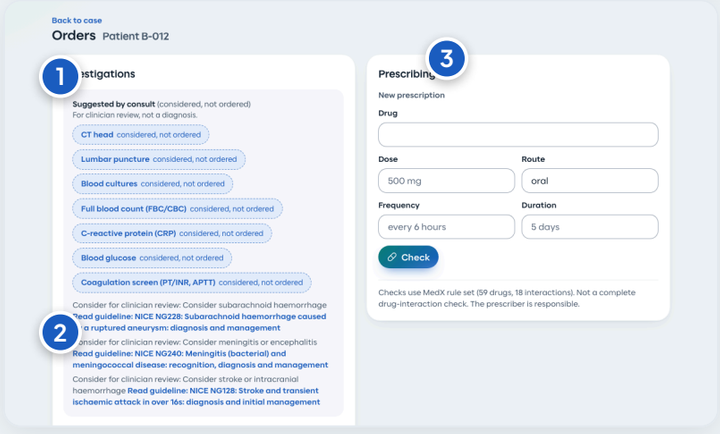
"Ask this next. Rule this out. You decide." Suggestions, never decisions: the nurse confirms every triage level.

Screens: /review//consult?mode=nurse and /vitals, synthetic case B-012, mock provider. Triage level is a NEWS2–based suggestion for adults (not pregnancy); the nurse confirms. docs/clinical/CONSULT-SPEC.md.

USE CASE: DOCTOR

The doctor gets a draft note and doctor-only considerations from a fixed list. Not a diagnosis.





1. **Draft, not a diagnosis**
   The note carries a banner: draft for clinician review. Nothing is final until signed.
2. **Signed note**
   Reviewer name and time are stored in the audit log; the patient's own name is redacted.
3. **Needs check**
   Anything without a verified quote is shown dashed as **Needs check**.
4. **Considerations**
   Doctor-only, from a fixed list of 12 presentations (not a full differential), each with a guideline link such as NICE NG228.
5. **Orders**
   Investigations are listed as considered, not ordered. Prescribing is checked by rules (next slide).

The consult assistant never diagnoses, orders, prescribes or decides urgency.

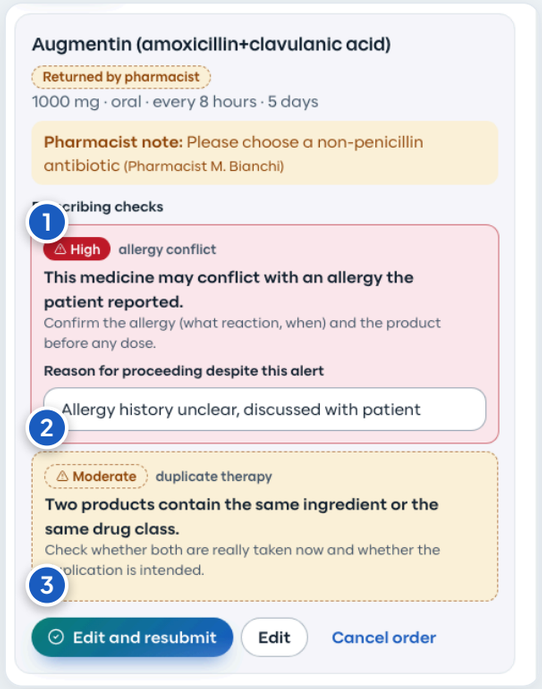


“Here is what to consider. What to do is up to you.” A fixed list with guideline links, not a model's guess.

Screens: /review//consult?mode=doctor and /orders, synthetic data. 12-presentation list: docs/clinical/CONSULT-SPEC.md; README limitations. Guideline names appear as shown in the app.

USE CASE: ORDERS WITH CHECKS

Prescriptions run through deterministic checks; overriding a high alert needs a reason.



CHECK	ALERT	EXAMPLE IN THE APP
Allergy conflict	High	Amoxicillin-clavulanate with a stated penicillin allergy
Interaction	High	From a list of 18 interactions
Duplicate therapy	Moderate	Two products with the same ingredient or class
Pregnancy	Check	Flags medicines to review in pregnancy
Dose range	Check	Adult dose ceilings for 31 drugs

RULE SET SIZE

59 / 18 / 31

drugs / interactions / dose ceilings in medrules.json (v3)

packages/shared/src/medrules.json; README

CALLOUTS

1 Alert with reason. **2** Prescriber note: "Allergy history unclear, discussed with patient". **3** Edit and resubmit, or cancel.

Limit: a small rule set. It misses what it does not contain. Alerts are code, not model output, and a model cannot clear one.

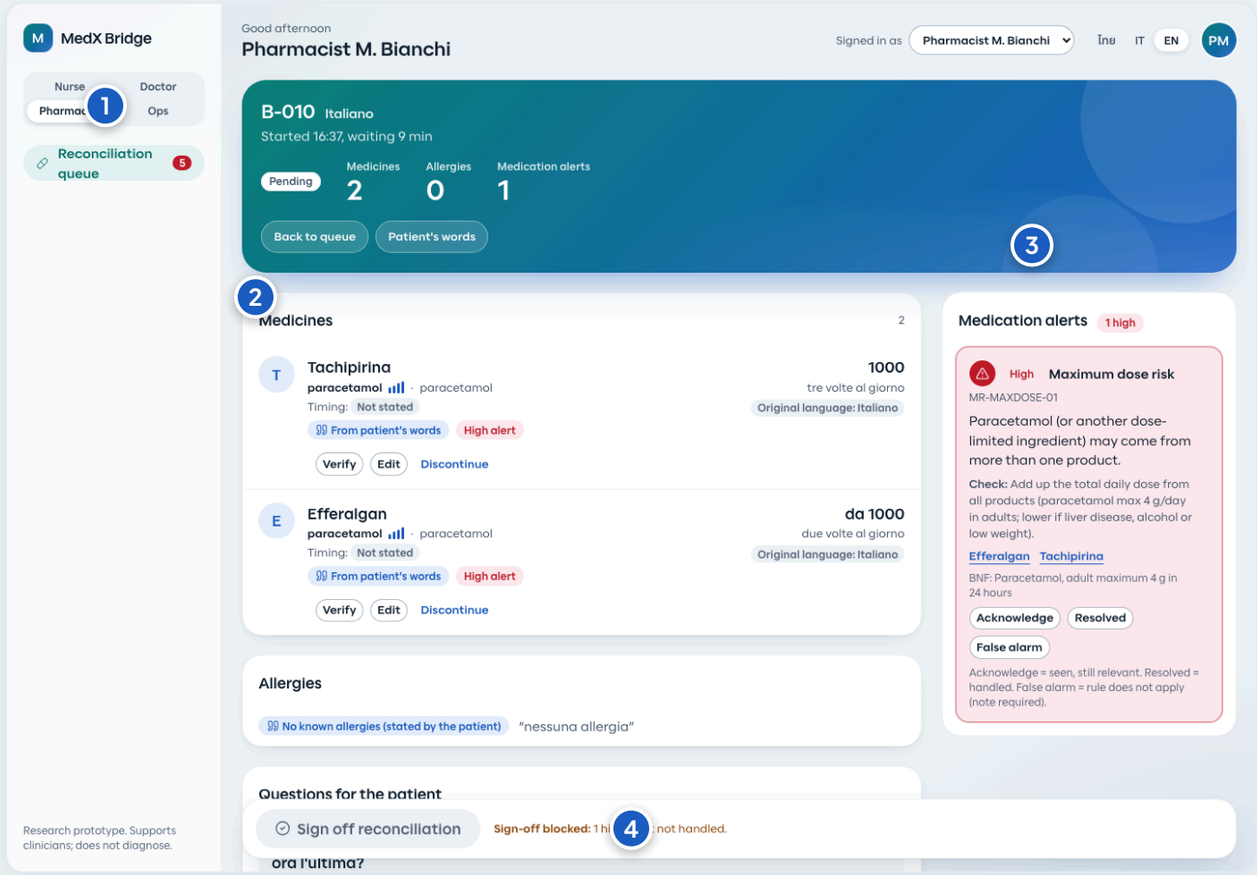


"Check first. Then prescribe." The alert is in code. The decision stays with the prescriber.

Screens: /review//orders and /pharmacy, synthetic data. Counts: medrules.json and README limitations (59 drugs, 18 interactions, dose ceilings for 31). Not a complete drug-interaction check.

USE CASE: PHARMACIST

The pharmacist sees what the patient said, with quotes, and verifies or returns each prescription.



1. **Case header**
   Language, medicines, allergies and alert counts at a glance.
2. **Medicines**
   Each row: patient's words in the original language, generic name, dose and timing, and status. Verify, edit, or discontinue.
3. **Medication alerts**
   High, moderate, or duplicate-paracetamol alerts with a short check to do. Acknowledge or mark false alarm.
4. **Sign off reconciliation**
   Blocked until every high alert is handled.

RETURN TO PRESCRIBER

A returned order carries the pharmacist's note back to the doctor, who can edit and resubmit or cancel.

The system never recommends a new drug, a dose, a substitution or stopping a drug.



"I verify. If something is wrong, I send it back with a reason." Sign-off stays blocked while a high alert is unhandled.

Screens: /pharmacy and /pharmacy/, synthetic case B-010, mock provider. Spec: docs/clinical/PHARMACY-SPEC.md.

SAFETY LAYER

What could go wrong, and what MedX does about it.

WHAT COULD GO WRONG	RISK	WHAT MEDX DOES
A warning sign is buried in a long answer	Urgent	18 deterministic red-flag rules run before any model call. A hit stops the flow and shows a red banner; no model output can clear it.
The model invents a symptom, medicine or allergy	Wrong fact	A field with no verified quote from the patient's own turns is **unsupported** and never shown as fact.
The model fails or is unsure	Guess	Provider error, schema error or low confidence means abstain: **needs human review**. A 40-second per-turn deadline.
Patient text leaks personal data	Privacy	Names, phones and IDs are redacted before storage or any model call. Synthetic data only.
A prescription conflicts with an allergy or another drug	Harm	Rules in code flag it. High alerts need a written override reason. Pharmacist verifies.
Someone acts on an unchecked output	Overtrust	Nothing is final until a clinician approves. Every action is in the append-only audit log with reviewer.
The model tries to add or remove flags	Control	The model's red-flag screen can only add flags, and those need clinician confirmation.



"I can be wrong. So the system is built to show its work." Seven risks, seven controls. Each one has automated tests.

Invariants and flow: docs/ARCHITECTURE.md (safety invariants table), apps/web/lib/pipeline/__tests__. Not clinically validated; the controls reduce risk and do not remove it.

RED FLAGS BEFORE ANY MODEL

Red-flag rules are code. They run first, and the model cannot clear them.

Patient turn
Text is redacted first

›

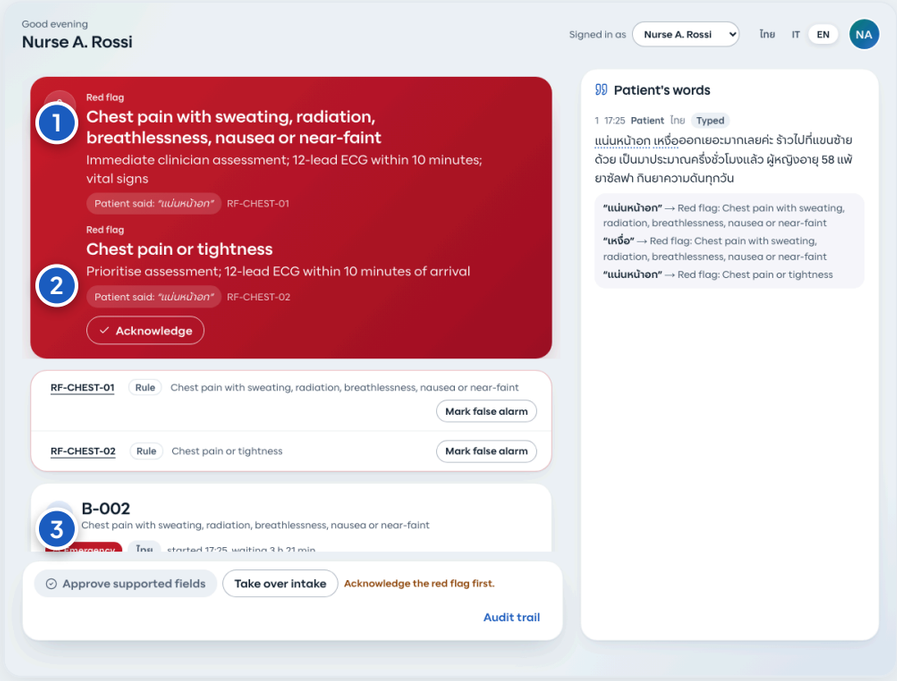
Red-flag rules
18 rules, TH / IT / EN, with negation cues

›

Stop or continue
Hit: status escalated, zero model calls

›

Model
Extraction and an add-only screen



1. **Red banner, urgency only**
   Chest pain with sweating, found by RF-CHEST-01, with the patient's quote shown in Thai.
2. **Evidence, not inference**
   The rule id and the quote that triggered it.
3. **Clinician actions**
   Acknowledge, mark false alarm, approve supported fields, or take over intake.

BLIND HOLDOUT, RULES ALONE

7/10

escalation cases caught; 3/8 false alarms. We publish the 3 misses: suicidal ideation, meningitis, seizure phrasing.

eval/results/holdout-rules-only.md; eval/REPORT.md (18 cases, authored blind)

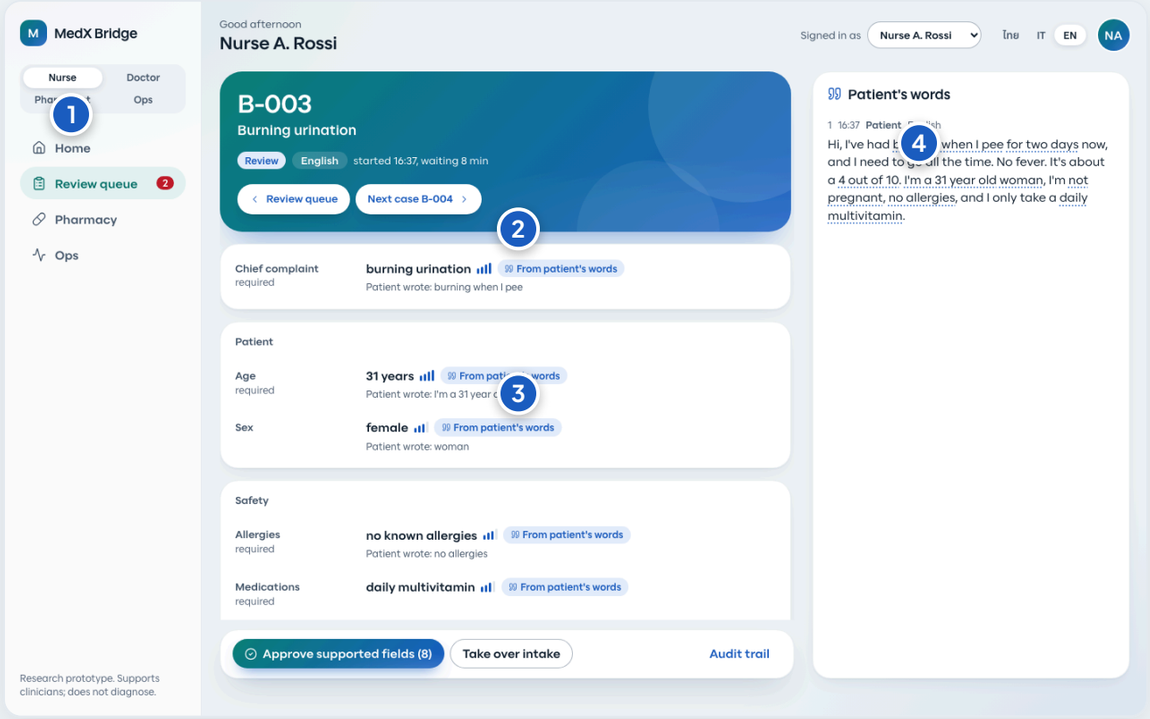


"I would rather ask a nurse than guess." Rules miss paraphrases. That is why a clinician reviews every flag and every case.

Rule count: packages/shared/src/redflags.json (18). Holdout: data/holdout/ (18 synthetic cases: 10 escalate, 8 not), rules v1.1 recall 7/10, false alarms 3/8, recorded 17 Oct 2026. A mock-provider arm adds no hits (labelled mock, not evidence).

EVIDENCE-LINKING

Every field points to the patient's own words. If it cannot, it says so.



1 Case header
Language, status, and a triage chip once vitals are confirmed.

2 From patient's words
Each field shows its quote on hover and links to the line in the transcript.

3 Missing or unverified
Required but absent fields are flagged for a follow-up question.

4 Patient's words panel
The original text, with the quoted passages highlighted.

Five statuses, always visually distinct

verified · unverified · missing · edited · approved

HOW IT IS CHECKED

Each model-proposed quote must be found in the patient's turns (fuzzy match across translation). If a majority of fields fail, the whole case abstains.

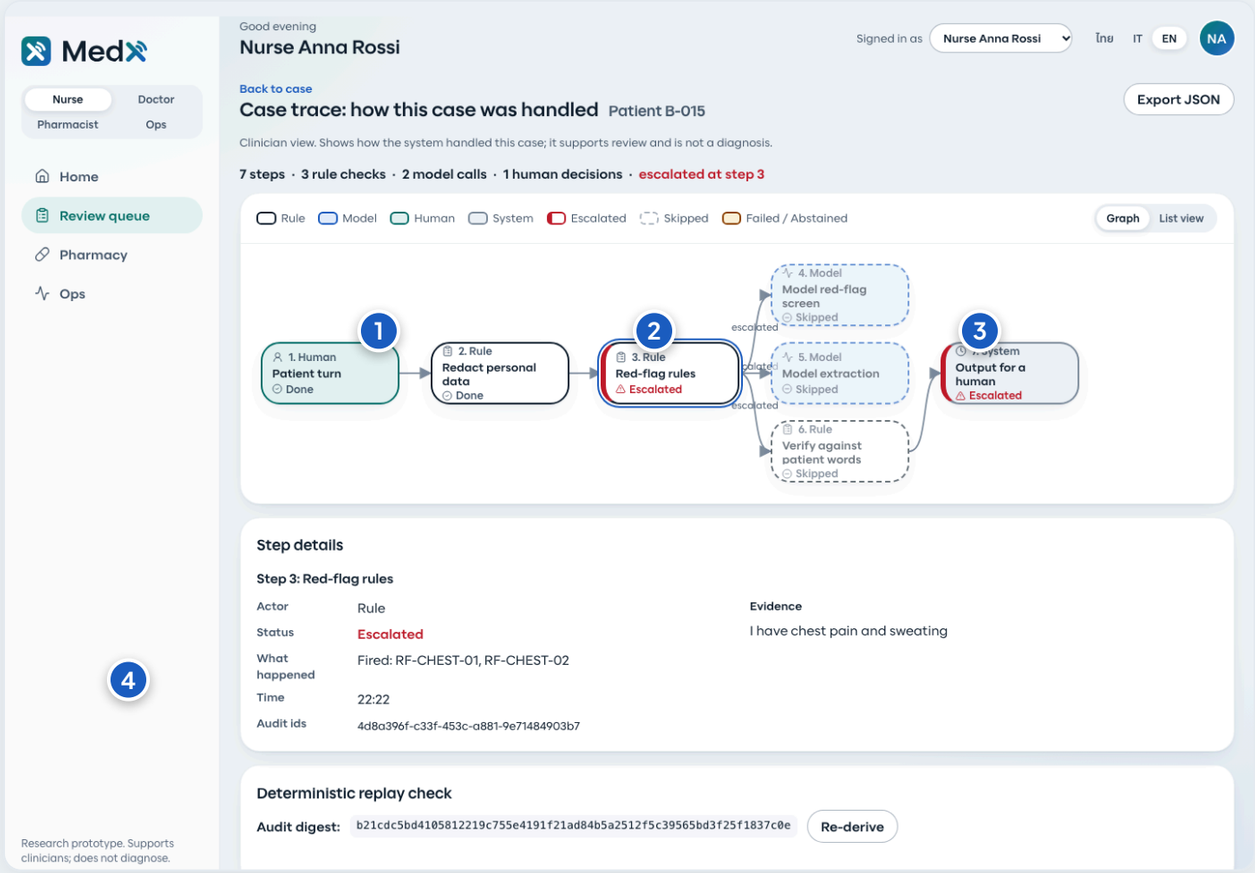


"Click a field. See the words." Evidence is the product; the summary is only the surface.

Screen: /review/, synthetic case B-003, mock provider. Verification logic: apps/web/lib/pipeline/verify.ts, docs/ARCHITECTURE.md step 5. Safety invariant 2 in CLAUDE.md.

CASE TRACE

Every case has a trace: the exact path of rules, model calls and human decisions.



1. **Graph of steps**
   Patient turn, redaction, red-flag rules, model steps, escalation, human decision.
2. **Model steps skipped**
   On a red flag the model steps are shown as skipped. Zero model calls.
3. **Step details**
   Actor, status, what happened and the rule id, with the audit id.
4. **Deterministic digest**
   A hash over the trace, and an Export JSON button. No personal data in the trace.

The trace is derived from the audit log. It answers "how was this case handled?" for a reviewer, a regulator, or a judge.

It describes what happened. It does not judge whether the decision was right.



"Here is how this case was handled, step by step." Rules, models and people are all in the same picture.

Screen: /review//trace, synthetic escalated case B-018, mock provider (docs/reports/screens/v11-trace-redflag.png). Case trace was added in waves 17–18 (docs/reports/w18-trace-ui.md).

ARCHITECTURE

One API gateway, one model client, and a mock fallback that is always labelled.

CLIENTS

Patient app
Expo / React Native, TH IT EN

Staff web
/review /pharmacy /ops, Next.js 15

›

API GATEWAY (THE ONLY BACKEND)

apps/web/app/api
Validate, redact, run pipeline, append audit. JSON store, single instance.

›

PIPELINE (NO UI IMPORTS)

redact -> red-flag rules -> extract + safety screen -> verify quotes -> abstain rules -> next question -> med alerts

STATUS OF THE AMD RUN

live

Live on the AMD developer API since 18 Oct. Accuracy and latency are measured in eval/REPORT.md (synthetic, small n).

docs/ARCHITECTURE.md; eval/REPORT.md header

MODEL CLIENT (LLM.TS)

OpenAI-compatible
chat/completions, JSON schema output, timeout, one repair retry, latency recorded per call.

›

EVALUATED ON

Gemma-4-31B-IT + DeepSeek-V4-Flash via the AMD developer API
Endpoint, model id and keys are env vars. Evaluated 18 Oct 2026 (synthetic, small n).

FALLBACK

Mock provider
Deterministic. UI and /api/health say mock, never AMD.

HONEST LABELS

If AMD_BASE_URL is unset, the provider is mock and the UI says so. Keys only from env vars; .env is never committed.

CONTRACTS

packages/shared: types, API shapes, tokens, redflags.json, medrules.json. Imported by web and mobile; changed only by the orchestrator.



“Same pipeline, any OpenAI-compatible model. The label never lies.”

Runs on AMD-hosted models through the AMD developer API (Gemma-4-31B-IT, DeepSeek-V4-Flash); measured on synthetic data, thresholds mostly not met.

docs/ARCHITECTURE.md (components, data flow, AMD integration). We do not claim that MedX runs on AMD Instinct or ROCm; latency and accuracy come only from eval/REPORT.md.

HOW WE BUILT IT

A human lead orchestrated AI agents in role lanes, and every lane was cross-reviewed.

ORCHESTRATOR

Claude Opus

Plans waves, owns shared contracts, reviews reports, integrates.

BUILDERS

Claude Sonnet

Web, mobile, pharmacist, clinician and ops screens; engine work in later waves.

ENGINE AND REVIEWER

OpenAI Codex

AI engineer lane (pipeline, API) and cross-review of UI work.

SPECIALISTS

Clinical, research, AI, design, pitch

Doctor rules, evidence, eval design, design system, deck.

REVIEW WAVE	REPORTS IN REPO	WHAT IT CHECKED
QA bug waves	7-qa-bugs, w14-qa-bugs	Flows, edge cases, regressions
UX audits and re-audit	w7-ux-audit, w8-ux-reaudit, w16-ux-final	Journeys, language leftovers, red-flag behaviour
Clinical-safety reviews	w9, w14 clinical-safety-review	Rules, alerts, claim boundary
Cross-vendor review	w6-engine-review, w5-codex-ui-review, w9-codex-final-review	Claude reviewed Codex's engine; Codex reviewed the UI

Every task ends with a report (status, files, tests run, evidence, risks, open questions). No report means not done; findings were fixed under regression tests.



“Many hands, one trail. Each wave left a report.” Humans decide; agents draft and review.

docs/reports/ (w1 to w19 role reports), docs/DECISIONS.md, README section Team MedX. Agent roles as described in the repo; no claim about code quality beyond the test suite and reviews listed.

CUSTOMER ANALYSIS

The buyer is a private clinic. The users are nurses, doctors and pharmacists. The patient benefits.

BUYER

Private hospitals and clinics with international patients, including tourist-area clinics.

Pains: interpreter cost and throughput, documentation time, verifiable AI, medication-error risk.

USERS

Nurses, doctors, pharmacists

Head nurse: documentation time. Doctors: want AI they can check. Quality team: medication errors.

BENEFICIARY

Patients with limited local-language skills

Tourists and migrant workers; Thai, Italian, English first.

STAKEHOLDER	JOB TO BE DONE	EVIDENCE IT MATTERS	VALIDATED WITH THEM?
CEO / CFO	Serve international patients without growing interpreter cost	Interpreter access is limited (37% in one European survey)	not yet
Head nurse	Cut documentation load	21% to 28% of shift on documentation (measured, ward)	not yet
Doctor	Trust a summary without re-asking	LLM omission/hallucination rates in studies	not yet
Quality team	Fewer medication-list errors	13% to 59% admission lists with an unintended error	not yet



“I do not claim customers. I claim a clear first customer profile.” Next step: a measured pilot at one clinic, with no savings figure promised.

Business case text: pitch/SUBMISSION.md. Evidence figures from docs/research (Carrasco-Sanz 2018; J Clin Nurs 2021; PHARMACY-PROBLEM.md). No customer interviews, pilots or agreements exist; none are claimed.

VALUE PROPOSITION

MedX is worth paying for if it makes safe intake faster to check, not faster to skip.

FOR THE CLINIC

- One intake in three languages without a custom integration per language
- Audit log and Case trace for every case
- Rules in code, so safety does not depend on a prompt

FOR THE CLINICIAN

- Every field links to the patient's sentence
- Unsupported fields are never shown as fact
- Suggestions with guideline links; you decide

FOR THE PATIENT

- Speak in their own language, confirm the transcript
- Red-flag screen: told to go to the desk now

PAIN	GAIN	MEASURE IN A PILOT
Interpreter wait	Intake starts before the interpreter is free	Time to structured summary
Re-asking	Doctor starts from linked quotes	Repeat questions per visit
Silent AI errors	Unsupported fields shown as such	Unsupported claims shown as fact (target: 0)
Medication list errors	Reconciled list with checks	Discrepancies found at pharmacist check

WHAT WE DO NOT PROMISE

No cost savings, no diagnosis, no triage decision. Gains are hypotheses until measured. Interpreters remain the gold standard where available (Karliner 2007).



“Faster to check, never faster to skip.” The gain is trust you can verify.

Value hypotheses are ours (assumption for discussion). Karliner LS et al., Health Serv Res 2007 (PMID 17362215). Pilot measures follow docs/ai/METRICS.md; the 0 target is safety invariant 2 in CLAUDE.md.

ALTERNATIVES AND POSITIONING

MedX does not replace interpreters or scribes. It structures the first words and keeps the trail.

ALTERNATIVE (CATEGORY)	WHAT IT DOES WELL	WHERE IT FALLS SHORT FOR INTAKE	MEDX DIFFERENCE
Professional interpreter, in person	Best communication quality (Karliner 2007)	Scarce, costly, a wait; not always available	Starts intake before the interpreter; hands over a structured case
Phone or video interpreting service	On-demand languages	Another step in the visit; not linked to the record	Capture and structure the patient's own answers asynchronously
Generic translation app	Fast, free, many languages	No clinical structure, no evidence trail, no red-flag rules	Fields with quotes, rules in code, audit log
Ambient scribe (category)	Drafts notes from the visit audio	Built for the doctor visit, not patient-side intake; fields may lack quotes	Evidence-linked fields; unsupported shown as unsupported
Paper or EHR intake forms	Already in use	One language; free text; no checks	Multilingual guided chat plus checks

Positioning. The safety-first intake layer for clinics with international patients: **evidence-linked, rule-checked, human-approved**.

We name categories, not products, because we have no verified product-by-product comparison. This table is our reading of the categories, not a benchmark.



“Not instead of an interpreter. Before one, and alongside one.” Honest about where each option wins.

Karliner LS et al., Health Serv Res 2007 (PMID 17362215) for interpreter quality. Other cells are our characterisation of categories (assumption for discussion); no competitor product, price or market share is stated.

REVENUE MODEL AND GO-TO-MARKET

Options under discussion, with no price or market size claimed.

REVENUE OPTIONS ASSUMPTION FOR DISCUSSION

OPTION	FIT
Per-site subscription	Simple for a clinic; one budget line
Per active clinician seat	Scales with use; needs seat management
Pilot fee, then subscription	Aligns with a measured pilot
Language packs / integrations	Later: add-ons per language or EHR

No price, margin or willingness-to-pay number has been validated. None is stated.

WHAT WOULD HAVE TO BE TRUE

Clinics accept a research-grade tool only after safety review, language review and legal review. The first revenue is not before those gates.

GO-TO-MARKET PHASES (PLAN, NOT COMMITMENTS)

Phase 0
Now: research prototype, synthetic data, public repo

›

Phase 1
Shadow review: de-identified clinic data with ethics approval; no patient impact

›

Phase 2
Supervised pilot at one clinic with clinician sign-off

›

Phase 3
More clinics and languages after review

FIRST CUSTOMERS TO APPROACH

Private clinics in tourist areas of Thailand with international patients. Italian-speaking tourists and Thai patients are the first two demo languages.

No conversation with any clinic has taken place. We are not claiming interest, pilots or signed deals.



“Earn trust first. Charge for it later.” Revenue follows the gates. It does not come before them.

Option list and phases are our plan (pitch/SUBMISSION.md business case, docs/PLAN.md). Everything on this slide is an assumption for discussion; there is no customer, pilot, price or market-size claim.

ROADMAP WITH GATES

No step towards clinical use without its gate: validation comes before expansion.

NOW

Hackathon prototype

4 role screens, 18 red-flag rules, med checks, Case trace, 2,500+ tests. Synthetic data, live on AMD-hosted models.

DONE

AMD run

Pre-declared eval (METRICS.md) run on the AMD developer API; published with unmet thresholds.

THEN

Shadow pilot

De-identified data, ethics approval, clinicians compare MedX to their own notes.

LATER

Supervised pilot

One clinic, sign-off on every case, measured outcomes.

GATE (ALL MUST PASS)	OWNER	STATUS
Native-speaker review of Thai and Italian wording	Language reviewers	open
Clinical safety review of rules, alerts and claims	Clinical lead	open: prototype review done, not clinical validation
Privacy: PDPA assessment, data-processing terms, hosting choice	Legal / DPO	open
Model results: holdout recall target 9/10, unsupported claims 5% or less	AI lead	measured 18 Oct: holdout 8/10, most thresholds not met
Regulatory position confirmed (software as medical device or not)	Regulatory adviser	open



“Gates are boring on purpose.” A gate that is not passed stops the roadmap.

Targets from docs/ai/METRICS.md (M11 holdout recall >= 9/10). Gate list and owners are our plan (assumption for discussion). docs/reports/w9 and w14 clinical-safety-review are internal prototype reviews, not clinical validation.

IMPACT: MEASURED TODAY

What we can verify today comes from the repo: tests, rules, data and a blind holdout.

AUTOMATED TESTS

1,872

pass in apps/web (vitest); regression tests for every fixed finding.

npm --prefix apps/web test, 18 Oct 2026

RED-FLAG RULES

18

deterministic rules in Thai, Italian and English, run before any model.

packages/shared/src/redflags.json

MED RULE SET

59 / 18 / 31

drugs / interactions / dose ceilings, adult only, small set.

packages/shared/src/medrules.json

SYNTHETIC DATA

52 + 18

cases (TH, IT, EN) and a blind holdout, never used to tune rules.

data/cases, data/holdout

BLIND HOLDOUT, RED-FLAG RULES ALONE (V1.1)

Emergency cases caught	7/10
False alarms on 8 non-urgent	3/8

Misses: HO-04 suicidal ideation, HO-08 meningitis, HO-09 seizure (paraphrased wording). False alarms: HO-11, HO-12, HO-17 (look-alikes). We report them because the holdout is the honest number.

eval/results/holdout-rules-only.md; eval/REPORT.md. 95% CI for 7/10 is 40 to 89%.

READ WITH CARE

eval/REPORT.md numbers come from the **real AMD-hosted models** on 40 synthetic cases and an 18-case holdout (re-run more than once): small n, wide confidence intervals, not clinical evidence.

NOT CLAIMED

No accuracy, time saving or patient outcome. Not clinically validated.



“Small, checkable numbers. No big claims.” Everything on this slide can be re-run from the repo.

Test count: vitest run, apps/web, 18 Oct 2026 (also README). Rules: redflags.json (18 rules). medrules.json v3 (59 drugs, 18 interactions); 31 dose ceilings per README limitations. Cases: data/cases (52 JSON), data/holdout (18 JSON). Holdout run recorded 17 Oct 2026, rules v1.1.

IMPACT: MEASURED AND TARGETS

The AMD-hosted Gemma run is in and misses most targets. The pilot is still ahead.

GEMMA-4-31B-IT + DEEPSEEK-V4-FLASH, AMD DEVELOPER API

METRIC	RESULT
Field precision and capture recall	88.1% (threshold 90%) and 26.2% (threshold 70%): not met
Turn latency p50 / p95	38.4 s / 42.2 s (target 6 s / 12 s): not met
Holdout recall, rules only vs rules + Gemma screen	7/10 vs 8/10 (false alarms 3/8, all from rules)
Unsupported claims with verifier	11.9% (threshold 5%): not met

From eval/REPORT.md (18 Oct 2026). We publish a result even if a threshold is not met.

PRE-DECLARED TARGETS TARGET

METRIC	TARGET
Holdout red-flag recall (rules + model)	≥ 9/10
Field capture recall	≥ 0.70
Unsupported claims with verifier	≤ 5%
Abstain recall on should-abstain cases	≥ 0.90
Unnecessary abstain rate	≤ 0.10
Turn latency p50 / p95	≤ 6 s / 12 s

Targets were written before any run (docs/ai/METRICS.md). They are goals, not claims.

PILOT MEASURES TO BE MEASURED

- Time from first word to structured summary
- Repeat questions per visit
- Discrepancies found at the pharmacist check
- Red flags caught vs missed, reviewed by clinicians
- Clinician trust and corrections per case

No target number is set for time saved. We will not state one before a measurement exists.



“Targets are promises to measure, not results.” Missing a target will be published as plainly as meeting one.

Thresholds: docs/ai/METRICS.md (M1, M3, M8, M11 and the abstain metrics). Live results: Gemma-4-31B-IT on AMD-hosted models, synthetic data, small n (eval/REPORT.md, 18 Oct 2026). We do not claim that MedX runs on AMD Instinct or ROCm.

RISKS AND MITIGATIONS

The biggest risks are clinical safety, privacy and over-trust. Each has an owner and a mitigation.

RISK	OWNER	MITIGATION TODAY	STILL OPEN
Clinical safety: missed red flag (rules miss paraphrases: 7/10 on holdout)	Clinical lead	Rules before any model; model can only add; clinician reviews; unit tests; holdout published	Broader clinician-authored rules; real-world validation
Wrong or invented fields	AI lead	Evidence quote required, else unsupported; abstain on low confidence	Measured: field thresholds not met; raise timeouts, improve prompts, re-run
Privacy and PDPA	Legal / DPO	Redaction before storage or model call; synthetic data only; keys in env	Redaction is imperfect; PDPA assessment; hosting region
Model provider dependency	Engineering	OpenAI-compatible client, env-driven endpoint, fallback model, labelled mock	Shared key: about 20 requests/min, 429 under concurrency
Language quality (Thai, Italian wording)	Language reviewers	Spot checks in UAT; follow-up questions in three languages	Native-speaker clinical review needed
Clinician over-trust or alert fatigue	Product	Status chips, red reserved for urgency, human approval for everything	Usability testing with real clinicians
Adoption	Business	Role-specific screens; no EHR change needed in the prototype	No clinic validated; EHR integration unknown



“I list what could go wrong before someone else does.” Mitigation reduces risk. It does not remove it.

Owners are role lanes, not named people. Mitigations: docs/ARCHITECTURE.md, README limitations, docs/reports/w9 and w14 clinical-safety reviews. Holdout: eval/results/holdout-rules-only.md. Reviewer identity is self-declared; the store is a single-instance JSON file.

FEASIBILITY AND LEGAL

A research prototype that supports clinicians. Not a medical device, no real patient data.

CLAIM BOUNDARY

- Supports clinicians; never diagnoses, treats, prescribes or decides urgency
- Language: "suggests for review", "flags for clinician", "decision support"
- Not clinically validated; not a medical device
- Synthetic data only

HUMAN IN THE LOOP

Nothing is final until a clinician approves. Overriding a high alert needs a written reason. Takeover is available at any point. Every action is in the audit log.

FEASIBILITY QUESTION	TODAY
Runs end to end?	Yes: 4 role screens, live on AMD-hosted models, Docker image
Model path?	OpenAI-compatible client; Gemma + DeepSeek on the AMD developer API (live)
Offline / low-cost?	Not tested; not claimed
Integrates with an EHR?	No; standalone for now
Legal basis for real data?	Not established; PDPA assessment needed

KNOWN LIMITS

- Doctor-only considerations come from a fixed 12-presentation list
- NEWS2 triage is for adults, not pregnancy; nurse confirms
- Prescribing checks use a small rule set
- Speech goes to a cloud speech service
- Reviewer identity is self-declared



"A tool that supports clinicians. Not one that replaces them." We stopped short of every claim we could not check.

CLAUDE.md claim boundary and safety invariants; README Limitations and safety; pitch/SUBMISSION.md. No regulatory classification has been obtained; that remains a gate.

WHY NOW AND THE ASK

Open models make a safety-first multilingual layer practical. We ask clinicians to test it.

WHY NOW

- Open models such as Gemma can run behind an OpenAI-compatible endpoint, so a clinic can choose where the model runs
- AMD Token Factory offers such an endpoint; we built to it and wait for the run
- Studies show LLM summaries need checks; our design answers with evidence-linking and rules in code

Gemma and Token Factory fit is a design choice. No claim about performance.

WHAT WE ASK FOR

1. **Clinicians and nurses**
   Review the screens and rules; tell us what is wrong.
2. **Language reviewers**
   Thai and Italian wording, clinic register.
3. **A clinic partner, later**
   Only for a shadow pilot after the gates.

NEXT 30 DAYS PLAN

- Re-run the eval at larger n when the shared endpoint is quiet
- Publish results, met or not
- Native-speaker review pass
- Safety review of rules with clinicians

Repo: github.com/KingPhuripol/MedX-AI. Live demo: medx.areazeroai.com.



“Help us break it, so a patient does not have to.” We ask for review, not for trust.

pitch/SUBMISSION.md; docs/ARCHITECTURE.md AMD integration. The plan is ours and not a commitment. No partner, pilot or agreement exists.

TEAM MEDX

A human lead orchestrating AI agents in role lanes, built for AMD Developer Hackathon ACT III.

LEAD

Human lead and orchestrator

Direction, decisions and review. Named team members: to be added.

CLINICAL AND RESEARCH

Doctor and medical researcher lanes

Red-flag and med rules, consult spec, cited problem evidence.

AI AND ENGINE

AI engineer and researcher lanes

Pipeline, eval design, pre-declared metrics.

PRODUCT

Design, web, mobile, pitch lanes

Design system, four role screens, patient app, deck.

HOW WE WORK

Role lanes with folder ownership; shared contracts changed only by the orchestrator; every task ends with a report; QA, UX and clinical-safety waves; cross-vendor review (Claude reviewed Codex, Codex reviewed the UI).

HONEST NOTE

The roles are AI agent lanes directed by a human; they are not employees or clinicians. We do not list named advisors or institutions because none are confirmed.



“Many lanes, one human who is accountable.” Thank you for reading the trail, not only the headline.

README Team MedX; CLAUDE.md layout and ownership; docs/reports/. Team member names are not listed here because they are not recorded in the repo; add from the lablab team page before sharing.

MedX

MedX: Multilingual AI Copilot for Clinic Visits



EVERY WORD, UNDERSTOOD. · A COPILOT SUPPORTS CLINICIANS AND NEVER DECIDES.

One patient journey, from the patient's own words to the pharmacy. Every field points back to what the patient said. People decide.

Repo: github.com/KingPhuripol/MedX-AI · Live demo: medx.areazeroai.com

Research prototype · synthetic data only · not a medical device · runs on AMD-hosted models through the AMD developer API (Gemma-4-31B-IT), results in eval/REPORT.md